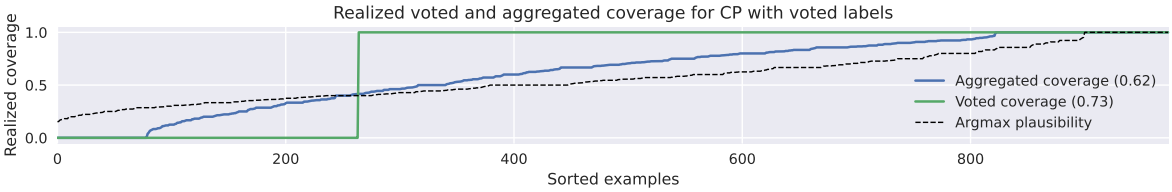

Figure 9: Realized voted coverage, i.e., $\mathbb{I}[\arg\max_k \lambda_{ik} \in C(X_i)]$ (green), and aggregated coverage, i.e., $\sum_{y\in[K]} \lambda_{iy}\mathbb{I}[y \in C(X_i)]$ (blue), for standard CP using voted labels, i.e., $\arg\max_k \lambda_{ik}$. Additionally, we plot the maximum plausibility per example (dashed black) as proxy of ambiguity. We sort examples on the x-axis according to each plot individually. Clearly, many cases are very ambiguous and aggregated coverage is underestimated severely (62% vs. the target of 73%).

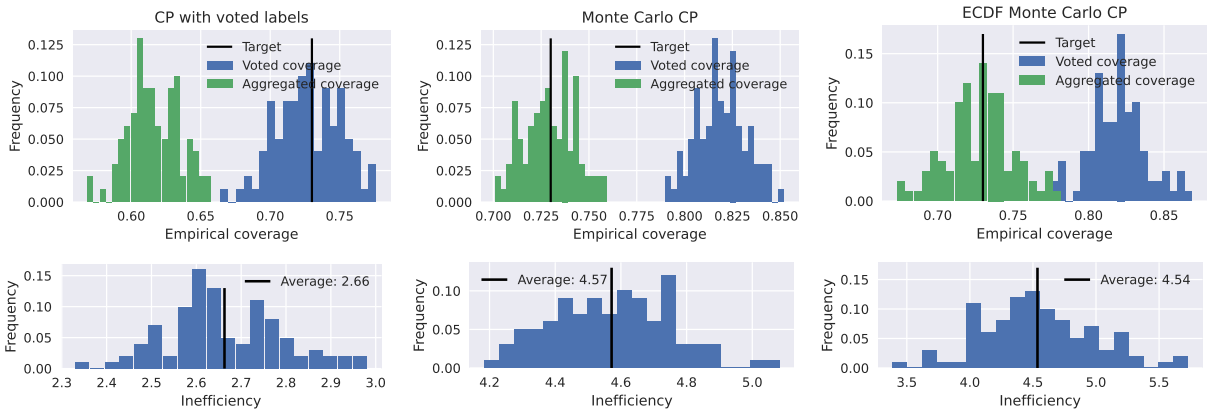

Figure 10: Comparison between CP applied to voted labels (left), Monte Carlo CP (middle) and ECDF Monte Carlo CP (right) in terms of voted coverage (blue) and aggregated coverage (green) across 100 random calibration/test splits (top) as well as inefficiency (bottom). We use $m = 10$. With voted labels does not reach the 73% aggregated target coverage (black). Monte Carlo CP overcomes this gap at the expense of higher inefficiency (bottom). Using ECDF-corrected p-values increases observed variation slightly due to the additional calibration data split (half of the original split).

against the voted label from the plausibilities[6]. We chose a coverage level of $1-\alpha = 73\%$ for our experiments (with results for $\alpha = 0.1$ in the appendix) to stay comparable to the base model.

In Figure 9, we highlight how ambiguous the plausibilities for skin condition classification are: in dotted black, we plot the largest plausibility against (sorted) examples. As baseline, we performed CP using the classifier's softmax output as conformity scores and calibrating against the voted labels $\hat{y} := \arg\max_k \lambda_{ik}$ per calibration example (X_i, λ_i). In blue, we plot the realized coverage by evaluating $\mathbb{I}[\hat{y} \in C(X_i)]$ per example. This is a step function and roughly 27% of the examples on the x-axis are covered. In green, we plot the realized aggregated coverage by evaluating $\sum_{k\in[K]} \lambda_{ik}\mathbb{I}[k \in C(X_i)]$ per example. For many examples, aggregated coverage lies in between $(0, 1)$ showing that the obtained prediction sets only cover part of the plausibility mass. More importantly, aggregated coverage is 62% on average, i.e., significantly under-estimated by calibrating against voted labels. This is the core problem we intend to address.

While the above results consider a fixed calibration/test split, Figure 10 shows our overall results across 100 random splits. As suggested above, on the left, we show how standard CP applied to voted labels does not achieve (as expected) the target of 73% for the aggregated coverage $\mathbb{P}_{\text{agg}}(Y \in C(X))$ (green). As highlighted in Figure 11, the prediction sets miss highly plausible conditions. In the middle and on the right, we demonstrate that Monte Carlo CP with $m = 10$ overcomes this problem and achieves (on average) the target aggregated coverage of 73%. Note that coverage w.r.t. the voted label (blue) increases alongside

[6]To be precise, in 72.6% of the cases, the voted label from the plausibilities is included in the top-3 prediction set derived from the predicted softmax output $\pi(x)$ *without* any conformal calibration.

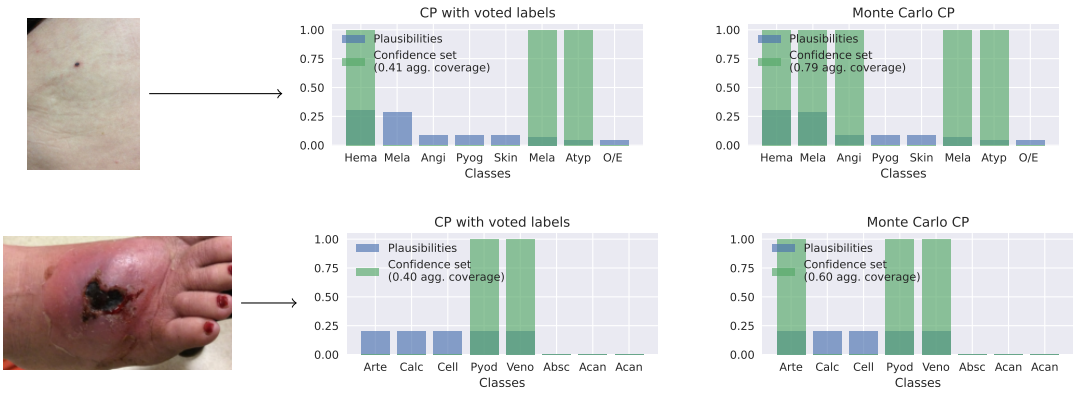

Figure 11: Comparison of CP with voted labels and Monte Carlo CP on two concrete examples. Both are ambiguous cases as shown by the high-entropy plausibilities. Monte Carlo CP clearly covers more plausibility mass (i.e., yields higher realized aggregated coverage), potentially improving patient outcome. Appendix C includes more qualitative results.

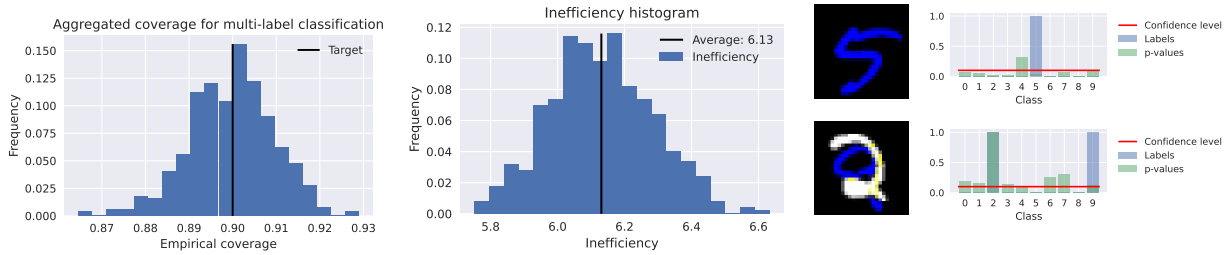

Figure 12: The multi-label CP strategy of Wang et al. (2015; 2014); Tsoumakas & Katakis (2007) is a slight variant of our Monte Carlo CP approach. As shown on this example of up to two overlaid, differently colored digits, Monte Carlo CP achieves target coverage of 90% (top left). However, it is free to decide how many labels to cover per examples (top middle) and, due to the poor performance of the base model (58.8% aggregated coverage), inefficiency is rather high. On the bottom, we show two examples of our dataset with the corresponding ground truth label set $\mathcal{Y}$ (blue) and the obtained Monte Carlo p-values (green).

aggregated coverage (i.e., the gap between voted and aggregated coverage remains approximately the same). In terms of inefficiency, avoiding over-confidence by improving aggregated coverage leads to a significant increase in the average prediction set size from 2.66 to 4.57. However, Figure 11 highlights that this is necessary for the prediction sets to include relevant conditions.

4.2 Case study: multi-label classification

In Figure 12 we consider a simple MNIST-based multi-label classification problem with up to two, differently colored digits per image. We trained 10 multi-layer perceptrons with 100 hidden units for each digit to determine if the digit is present in the image. This simple classifier achieves 58.8% aggregated coverage when thresholding the 10 individual sigmoids at 0.5. As discussed in Section 3.6, a common strategy (Wang et al., 2015; 2014; Tsoumakas & Katakis, 2007) of performing multi-label CP consists in repeating each example according to its number of labels (here, at most 2). We can achieve something similar with rigorous aggregated coverage guarantee by uniformly sampling labels to perform Monte Carlo CP. We illustrate this procedure in Figure 12 (top left) where we show that it achieves the 90% coverage target. Furthermore, our discussion in Section 3.2 establishes the corresponding guarantee of $1 - 2\alpha$ without ECDF correction. However, it is important to understand what aggregated coverage means for multi-label classification: CP decides how many of the labels it intends to cover per example in order to achieve the marginal coverage guarantee. This is highlighted in Figure 12 (bottom) showing the corresponding p-values for two examples. In the first example, only one out of two examples is covered by the prediction set.

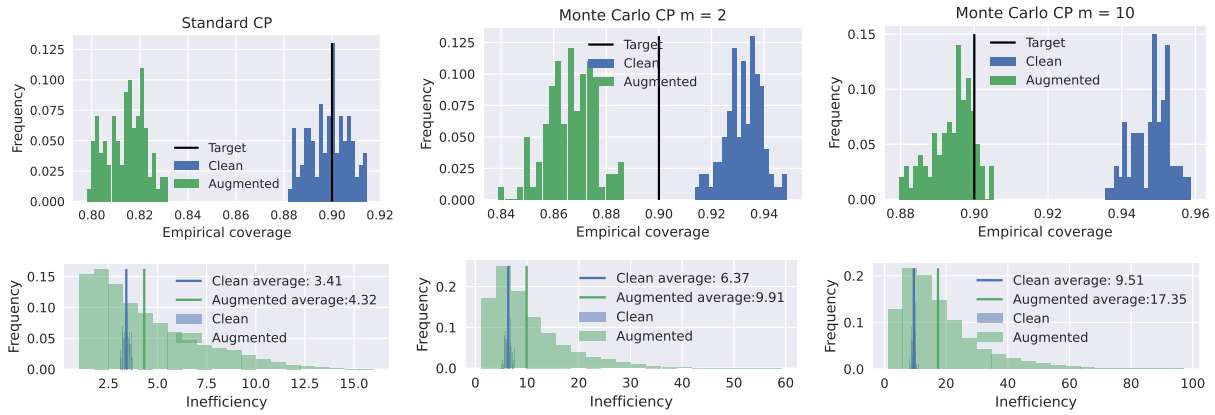

Figure 13: Standard CP applied to original images (left) and Monte Carlo CP (middle and right) applied to the original plus augmented images from ImageNet for $m = 2$ (middle) and $m = 10$ (left). We show coverage on original (blue) and augmented images (green, average across 25 AutoAugment augmentations per image). Monte Carlo CP is able to increase coverage on augmented images significantly at the expense of higher inefficiency (on both original and augmented images). Using more augmentations during calibration generally increases coverage on augmented images and reduces the observed variation across random calibration/test splits.

4.3 Case study: data augmentation and robustness

We also apply Monte Carlo CP in the data augmentation and robustness setting outlined Section 3.6. Specifically, we took a pre-trained MobileNet V2 (Howard et al., 2017) achieving 71.3% (top-1) accuracy on the first 5k test examples of ImageNet (Russakovsky et al., 2015) and additionally evaluated it on augmented images using AutoAugment (Cubuk et al., 2018). We generated 25 random augmentations per test example. The model achieves 60.2% accuracy on average, significantly lower than on the original images. Similarly, Figure 13 shows a significant gap in coverage when calibrating only on the original images. While coverage on original examples is around the target of 90%, depending on the random calibration/test split, coverage of augmented images is only slightly above 80%. With the Monte Carlo CP procedure described in Section 3.6, we can use as calibration data the original *and* augmented images. This procedure increases coverage on the augmented images significantly at the cost of higher inefficiency. As coverage is marginal and thus split across augmented and original images, the coverage increase is largest when using more augmented images during calibration. We interpret these results in two ways: First, there is no reason anymore to train state-of-the-art models with data augmentation but discard augmented images during calibration. Second, our Monte Carlo CP approach is effective in improving robustness against augmentations or other corruptions.

5 Discussion

In many classification tasks, the available ground truth labels arise from a *voting* process relying on several expert annotations resulting in a one-hot distribution $\mathbb{P}^{Y|X}_{\text{vote}}$. However, in scenarios where annotators tend to disagree because ground truth labels are ambiguous, this voting approach ignores the label uncertainty. Thus, performing CP with such voted labels can only guarantee coverage w.r.t. $\mathbb{P}^{Y|X}_{\text{vote}}$. This can have severe consequences, especially in safety-critical applications such as the dermatology case study in this paper. Instead, we use standard procedures from the literature to *aggregate* expert opinions and return a non-degenerate conditional probability distribution $\mathbb{P}^{Y|X}_{\text{agg}}$ that can capture uncertainty about Y. In this paper, we proposed two Monte Carlo CP procedures which can output prediction sets satisfying some pre-specified coverage guarantees under the corresponding distribution $\mathbb{P}_{\text{agg}} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}_{\text{agg}}$ by sampling m exchangeable synthetic labels from $\mathbb{P}^{Y|X}_{\text{agg}}$ for each calibration data. This allows to reduce the variability of the empirical coverage across realizations of the sampled labels. For a test example X, the prediction set outputted by such procedures can be thought as the one expert would assign if they had access to X and their annotations